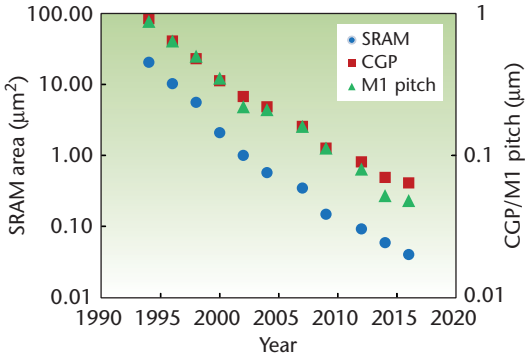


Figure 2. Three key measures of integration density as a function of time. Blue dots show static random access memory (SRAM) density. Green triangles show M1 pitch, the minimum wire-to-wire spacing in the first wiring layer. Red squares show CGP or contacted gate pitch, the minimum spacing between transistors. Progress is still rapid, but there's evidence of slowing in recent years. Data compiled from published literature by Chi-Shuen Lee at Stanford University during his PhD thesis research with one of the authors (Wong). The data is available at https://purl.stanford.edu/gc095kp2609. Current, up-to-date data can be accessed at https://nano.stanford.edu/cmos-technology-scaling-trend.

performance gains have been muted. Furthermore, the more straightforward elaborations of the standard von Neumann computer architecture have already been implemented,[7] and prospects for significant performance gains from further increases in parallelism appear limited even at the multicore level.[14,15] It's no surprise then that the replacement cycle for computing equipment of all sorts has lengthened. An increasing number of semiconductor manufacturers are finding profits by investing in development of product attributes, such as architectures for improved memory access, that have little to do with smaller feature size.

Figure 2 shows progress in three key indicators of the achievable integration density for complex digital logic circuits since 1994. Progress is still rapid, but taken together, these indicators show a significant slowing in the last decade. The Moore's law development cadence—the regular doubling of integration density—appears to be slowing.

While end users still see improvements in performance and energy efficiency as greater numbers of memory devices are brought closer to processor cores, these gains cannot be long sustained with current memory devices and architectures. Static random access memory (SRAM), the fast memory that now occupies over half the surface area of a microprocessor chip, uses six FETs to store a bit of information. Dynamic random access memory (DRAM), the slower but denser and therefore less expensive memory that usually resides on memory chips peripheral to the processor, uses one FET and one capacitor to store a bit. Flash memory, the very dense but rather slow memory that stores data when the power is off, uses one FET with a specially designed gate structure to store one bit (or more recently, several bits). Thus each of these dominant devices in today's memory hierarchy is subject to scaling constraints similar to those for the FETs used in logic, plus additional constraints unique to each device.

Despite these daunting problems, we're very optimistic about the prospects for further dramatic advances in computing technology. After decades of progress centered on miniaturization of the CMOS transistor, we see a growing potential for advances based on the discovery and implementation of truly new devices, integration processes, and architectures for computing. By truly new devices, we mean devices that operate by physical principles that are fundamentally different from the operating principle of the FET and are therefore not subject to its fundamental limits, particularly the voltage scaling limit. By truly new integration technologies, we mean monolithic integration in three dimensions in a fine-grained manner that immerses memory within computational units. And by truly new architectures, we mean circuit- and higher-level architectures that are much more energy efficient than the von Neumann architecture, particularly for the important algorithms and applications of the coming decades. We now touch briefly on some of the emerging research concepts that fuel our optimism.

New Devices for Logic

As we write, several distinct physical principles are known by which a voltage-gated switch (that is, a transistor-like device) might avoid the voltage scaling limit of the conventional FET.[8,16] For example, some of these operating principles invoke a physical mechanism that breaks the direct link between externally applied operating voltage and the internal potential that gates the flow of current. Of course, just changing the device physics to embody one of these operating principles doesn't guarantee a more energy-efficient low-voltage digital switch. For each proposed device concept, the switching characteristics and other important device attributes will depend critically on the achievable properties of materials and the details of the device structure.